In-Band Lattice Zero-Knowledge Proofs for Compressed-Domain H.264 Video Steganography

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***Abstract*—Proof-carrying steganography hides a payload in a cover medium together with a zero-knowledge proof about it, so that a verifier can check a claim from the medium alone. Existing systems either rely on pairing- or discrete-logarithm-based SNARKs, which do not survive quantum adversaries, or deliver the proof beside the medium, where it is easily stripped or swapped. We present a system that embeds a lattice zero-knowledge proof directly in the CAVLC residuals of an H.264 Baseline bitstream and verifies it from the received video, without side files, metadata boxes or access to the cover. The proof is a Fiat–Shamir-with-aborts proof of knowledge of an opening of a BDLOP commitment, instantiated at an estimated 138–140-bit classical (125–127-bit quantum) core-SVP level. Its statement binds a carrier-normalized digest of the video, the verifier-recomputed carrier positions, a one-use verifier challenge and an issuer-signed relation registry; carrier normalization breaks the circularity between a proof and the video that contains it. Two modes are supported: a 7 589-byte envelope proving possession of a registered credential, and a 14 309-byte envelope committing to a hidden payload that can later be disclosed or linked to another commitment. On standard CIF and 4CIF sequences every completed run verified from the stego video alone and, where tested, rejected replayed sessions and single-bit tampering; luma PSNR on modified frames is 40.0–45.4 dB. Targeted classical steganalysis does not separate cover and stego, whereas a learned per-frame detector reaches a detection error of 0.40 across clips, which we report as a limitation. We state precisely what the proof does and does not establish.**

***Index Terms*—Video steganography, H.264/AVC, CAVLC, zero-knowledge proofs, lattice-based cryptography, post-quantum cryptography, steganalysis.**

I. Introduction

COMPRESSED-DOMAIN video steganography modifies the syntax elements of an encoded stream, such as quantized transform coefficients or their entropy codes, instead of decoding and re-encoding pixels. It keeps the stream standard-compliant, avoids re-compression artefacts and, for H.264/AVC, has a long literature [1]–[3]. A recent line of work attaches cryptographic evidence to media: a zero-knowledge proof that a hidden or withheld value satisfies a relation, so that a verifier learns that a claim holds without learning the value [4]–[9]. Combining the two gives *proof-carrying steganography*: the medium carries a hidden envelope that a verifier can check from the medium alone.

Two problems remain open for video. First, the proof systems used in this setting are pre-quantum: pairing-based SNARKs such as Groth16 [10], or discrete-logarithm-based folding schemes. Their proofs are small and easy to hide, but their soundness and the hiding of the commitments they prove about rest on discrete-logarithm-type assumptions that a large quantum computer breaks. Media evidence is often expected to remain checkable for years, so it is worth asking what a post-quantum alternative costs in a channel whose capacity is measured in kilobits per hundred frames.

Second, where the proof lives matters. A proof in a side file or a metadata box (as in content-credential manifests [11]) is removed by any re-muxing tool and can be swapped between videos. Moving the proof into the bitstream creates a circularity: the proof should bind the video, yet it cannot depend on a hash of a bitstream that contains the proof itself.

This paper addresses both problems for H.264 Baseline (CAVLC) video. Our contributions are:

- **An in-band design.** A lattice zero-knowledge proof travels inside H.264 CAVLC residuals and is verified from the stego video alone, with no side file, SEI message or cover. A *carrier-normalized* video commitment, computed with every carrier coefficient replaced by a canonical value and with carrier positions fixed from a placeholder envelope of the final length, removes the proof/video circularity (Section V).
- **A verifier-built statement.** The Fiat–Shamir statement binds the recomputed video commitment, the recomputed carrier positions, a one-use verifier challenge, and a relation registry signed by an issuer with ML-DSA-65 [12]. The verifier never reads a statement field from the video, which gives replay protection and anchors trust at the verifier.
- **Two proof modes and linkage.** A registered mode proves possession of an issuer-registered credential (7 589-byte envelope), and an in-band mode commits to a hidden payload of up to 766 bytes (14 309-byte envelope) that can later be disclosed or linked, by an equality proof, to another video or to a credential without being revealed.
- **A concrete, reproducible instantiation and evaluation.** We give a parameter set with core-SVP estimates and known-answer vectors, and we evaluate capacity, end-to-end embedding and video-only verification, decoded quality, classical and learned steganalysis, the choice of carrier, and the cost relative to Groth16, ML-DSA and a lattice zkSNARK (Sections VII–VIII).

We are explicit about the limits (Section IX): soundness is relaxed, the security argument is in the classical random-oracle model, the implementation is neither audited nor constant time, and a learned per-frame detector does separate cover and stego better than chance.

II. Related work

A. Proof-carrying and provenance media

PhotoProof [4] introduced cryptographic image authentication that survives a set of permissible transformations, using proof-carrying data over a signed original. ZK-IMG [5], VerITAS [6] and VIMz [7] prove image transformations with SNARKs, lookup arguments or folding schemes, and Eva [8] proves the authenticity of lossily encoded (e.g. H.264) videos with folding-based incrementally verifiable computation and a 448-byte proof. These systems target provenance of edits from an attested capture; the proof is an object delivered with the medium, and their assumptions are pre-quantum. Content credentials [11] attach signed provenance manifests as metadata, which survive only as long as the container keeps them. Closest to us in placement is ZK-WAGON [9], which embeds a zk-SNARK about an image-generation model into the generated image by spatial-domain LSB steganography; it is image-domain, embeds a gzip-compressed Halo2 proof of about 100 kB in the pixel LSBs, and relies on a pre-quantum SNARK. Adelsbach and Sadeghi [13] proved the presence of a watermark in zero knowledge, which is the converse problem (a proof *about* hidden data, kept outside it). To our knowledge no prior system carries a lattice zero-knowledge proof inside a compressed video bitstream and verifies it from the stream alone; we make no provenance claim of the kind PhotoProof or Eva make.

B. Compressed-domain H.264 steganography and steganalysis

Surveys [1]–[3] classify H.264 schemes by the syntax they modify: intra-prediction modes, motion vectors, quantized DCT coefficients and the entropy-coding layer. Coefficient schemes must control *intra-frame distortion drift*, the propagation of a modified block into intra-predicted neighbours [14]. Within CAVLC, the signs of trailing ± 1 coefficients ("trailing ones") can be flipped without changing the code length [15]; we compare this carrier with the magnitude-parity carrier we adopt (Section VIII-E). Video steganalysis [16] has concentrated on motion-vector embedding [17]; for all-intra streams the natural attacks are image-steganalysis features applied to decoded frames, such as spatial rich models with ensemble classifiers [18], [19] and CNN detectors [20], together with targeted histogram attacks on the modified domain [21], [22]. Evidence pooled over many frames of one video is stronger than per-frame evidence [23], a point we return to in Section IX.

C. Lattice zero-knowledge proofs

BDLOP commitments [24] with Fiat–Shamir-with-aborts openings [25], [26] are the basis of the exact and linear/product proofs of Attema et al. [27], Esgin et al. [28] and Lyubashevsky et al. [29], implemented in LaZer [30]; ML-DSA [12], [31] uses the same masking-and-rejection technique for signatures. Succinct lattice arguments, including LaBRADOR [32], LatticeFold [33], RoKoko [34] and SALSAA [35], target proof size for large relations; the versions we could evaluate disable zero knowledge in their reported configurations, prove knowledge soundness only, or (for folding) do not provide a standalone zero-knowledge prover. Designated-verifier lattice zkSNARKs [36], [37] are compact, but the oblivious LWE-sampling assumption used in the analysis of the standard-model constructions, including ISW21, is invalidated against quantum adversaries [38] (without an attack on the constructions themselves). We evaluated these candidates as backends against four gates (zero knowledge, expressibility of the verifier-built statement, runnability on the target host, and fit in the measured carrier capacity); none passed all four, which motivates the small, self-contained BDLOP opening proof used here. The relation we need, knowledge of one commitment opening bound to a statement, is exactly the one the BDLOP opening proof covers.

III. Background

a) H.264 CAVLC: In the Baseline profile [39], [40], the residual of each 4×4 block is entropy-coded with CAVLC: a `coeff_token` giving the number of non-zero coefficients and of trailing ones (up to three final ± 1 values in reverse zig-zag order), one raw sign bit per trailing one, variable-length level codes for the remaining coefficients with an adaptive suffix length, and run-length codes for the zeros. The context of `coeff_token` depends on the number of non-zero coefficients in neighbouring blocks, so a change that alters that number propagates to the coding of later blocks. In intra slices each block is also a prediction reference for its neighbours, which is the origin of intra drift.

b) BDLOP commitments: Let $\mathcal{R}_q = \mathbb{Z}_q[X]/(X^N + 1)$ with N a power of two and $q \equiv 5 \pmod 8$ prime, so that $X^N + 1$ splits into two irreducible factors and every non-zero element of small norm, in particular every difference of two challenges, is invertible [41]. With public $B_1 = [I_n \,|\, B_1'] \in \mathcal{R}_q^{n \times k}$, $b_2 = [0^n \,|\, 1 \,|\, b_2'] \in \mathcal{R}_q^{1 \times k}$ and $k = n + \lambda + 1$, a commitment to $m \in \mathcal{R}_q$ with ternary randomness r is $t_1 = B_1 r$, $t_2 = \langle b_2, r \rangle + m$. It is computationally hiding under Module-LWE and binding under Module-SIS [24], [42].

c) Fiat–Shamir with aborts: The opening proof is a Σ-protocol in which the prover masks the response $z = y + cr$ with a uniform y and aborts unless z lies in a range that makes its distribution independent of r [25], [26]. It is made non-interactive with the Fiat–Shamir transform [43]. The original reprogramming argument for this transform had a gap that was fixed by Barbosa et al. [44] and Devevey et al. [45]; quantum-random-oracle results for Fiat–Shamir [46], [47] are generic and not tight for our setting.

IV. System and threat model

a) Roles: An *issuer* publishes a relation registry signed with ML-DSA-65. Each registry record pins a proof suite, the hash of the verification key (including, in registered mode, the holder's credential commitment), the parameter-set identifier and a policy hash over the codec policy and the carrier profile; the registry has an epoch. A *holder* owns a payload or a credential and produces the stego video. A *verifier* issues one-use session challenges, receives videos and

decides. The *adversary* observes all videos and challenges, replays, edits, re-encodes and re-embeds videos, runs its own holder instances, and performs steganalysis.

b) Trust: The verifier trusts only (i) the issuer public key, (ii) a minimum registry epoch (its rollback floor), (iii) the parameter set and carrier profile it is configured with, and (iv) its own session store. Nothing read from the video is trusted before verification succeeds. The holder and verifier share a symmetric carrier-ordering (stego) key when undetectability is required; the key can instead be derived from public configuration (Section V-G).

c) Goals: (G1) *Soundness bound to video and session*: if the verifier accepts, then whoever produced the video knew a (relaxed) opening of the bound commitment, for exactly this carrier-normalized video, carrier layout, registry entry and session. (G2) *Zero knowledge*: the video reveals nothing about the payload or the opening beyond the statement. (G3) *Integrity*: any change to non-carrier syntax, and any change to the envelope bytes carried by the carriers, is rejected. (Changes to a few carrier bits that hold chunk-framing fields are currently not all rejected; see Section IX.) (G4) *Replay resistance*: an envelope is accepted at most once, and only for a challenge the verifier issued and has not consumed. (G5) *Linkage and disclosure*: a holder can later show that two commitments hide the same payload, or open a commitment to a chosen party. (G6) *Undetectability* against an adversary without the stego key, measured empirically by steganalysis.

d) Non-goals: The proof does not establish that the payload is true, that a camera recorded the video, that an encoder ran correctly, or who the creator is in in-band mode (anyone holding a session challenge can produce an in-band proof about a fresh commitment; identity requires registered mode). The channel is fragile by design: re-encoding or transcoding destroys the envelope, and verification then fails closed. Robustness is not a goal.

V. Construction

Figure 1 summarizes the embed and verify flows. The proof system is specified in Section V-D; the remaining subsections describe how its statement is tied to the H.264 stream.

A. Carriers

In every eligible luma 4×4 block of an IDR slice, the carrier is the first AC coefficient in zig-zag order with $|level| \geq 4$ that is not a trailing one. One bit is written into the least-significant bit of its magnitude by a change of ±1, keeping the sign; the coefficient stays eligible and, because the selection rule is also required to be *rank-stable* (no earlier candidate becomes eligible after the flip), a blind extractor that applies the same rule to the stego stream finds the same position. A block is used only if its re-encoding with the modified carrier round-trips through the CAVLC parser of the rewritten slice (*patchability* check), and at most one coefficient per block is modified. The embedder writes the bits by re-encoding the affected blocks and rewriting the slice payload, which may change its length. An alternative profile uses the sign of a trailing one as the carrier (Section VIII-E).

B. Carrier layout and the carrier-normalized commitment

The stream is divided into segments of $F = 100$ frames. The envelope is framed into chunks of a fixed size (a 23-byte header with magic, version, index, count, total and chunk sizes and a CRC-32 of the chunk body) and one chunk is embedded per segment. Within a segment, carrier positions are ordered by a keyed permutation; the ordering key is derived with HMAC-SHA-256 from the stego key. Let P be the ordered list of global carrier positions used by all chunks.

The circularity is broken by fixing everything the statement depends on *before* the proof exists. The embedder first frames a *placeholder* envelope of exactly the final length; this determines the chunk sizes and hence P. It then computes, for each segment, the SHA-256 digest of the segment bitstream with every carrier coefficient in P replaced by a canonical value (magnitude with its least-significant bit cleared, sign kept; a sign carrier is set to $+1$), and combines the segment digests into D with a domain-separated hash. The video commitment is

$$V = \text{SHA-256}(dom_V \,\|\, D),$$

with the domain tag dom_V = zkstego/video-zk/video-commitment/v1. After proving, the embedder checks that the real envelope produces exactly the placeholder's chunk layout and aborts otherwise. Because D normalizes exactly the positions that carry the envelope, it is the same before and after embedding; any change outside the carriers changes D, and any change of an envelope byte invalidates the proof or the statement. The framing fields themselves are a partial exception (Section IX).

C. Statement

The Fiat–Shamir statement x is the fixed-width (274-byte) encoding

$$x = ver \,\|\, rel \,\|\, pid \,\|\, root \,\|\, epoch \,\|\, N_s \,\| \\ codec \,\|\, profile \,\|\, |P| \,\|\, \mathsf{H}(P) \,\|\, V,$$

with a 16-bit version, 64-bit epoch and carrier count, and 32-byte values otherwise: relation identifier, parameter-set identifier, registry root, session challenge, codec-policy identifier, carrier-profile identifier (a hash of the carrier contract, segment length, chunk size, envelope length and mode), carrier-position hash and video commitment. The verifier builds x only from its pinned configuration and from values it recomputes from the received stream.

D. Proof system

Algorithm 1 gives the opening proof and its verification for statement x and commitment $t = (t_1, t_2)$; H is SHAKE-256 with domain separation and length prefixes, and $\mathsf{SampleInBall}$ is the ML-DSA-style sampler that outputs $c \in \mathcal{R}_q$ with exactly κ coefficients in $\{\pm1\}$ and the rest zero. The masks are derived deterministically from fresh entropy, the statement, the commitment and the witness (hedged), so a mask is never reused across statements. Nothing from a rejected attempt leaves the prover. The equality proof used for linkage runs

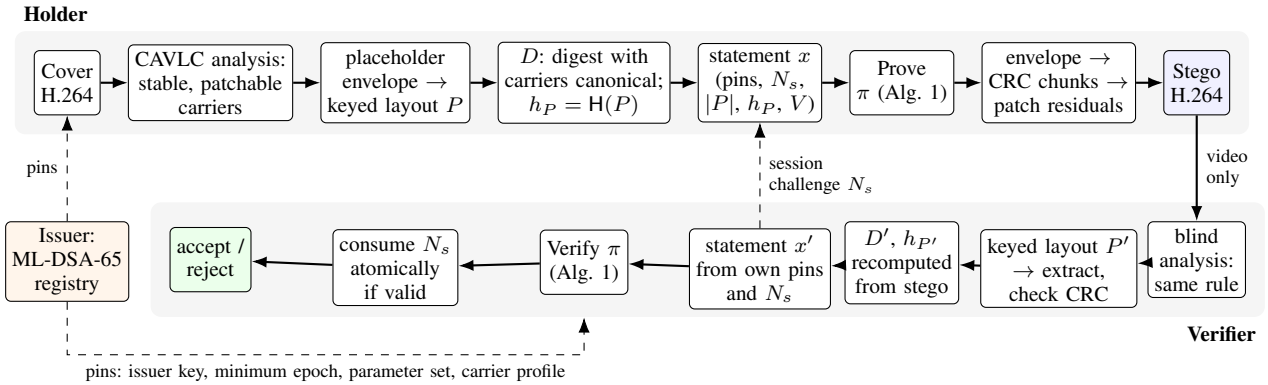


Fig. 1. Embedding (top) and video-only verification (bottom). Carrier positions and the carrier-normalized digest D are fixed from a placeholder envelope of the final length before the proof is computed, so the statement can bind them without depending on the proof bits. The verifier recomputes every statement field from the stego video and its own pinned configuration.

Algorithm 1 Opening proof (prover) and verification.

Prove(pid, x, t, r) ▷ $B = \gamma - \kappa - 1$
require $\|r\|_\infty \le 1$ and $B_1 r = t_1$
$K \leftarrow \text{SHA3-256}(\mathit{entropy} \,\|\, x \,\|\, t \,\|\, r)$
for $i = 1, \ldots, 512$ **do**
  $y \leftarrow$ uniform in $[-\gamma+1, \gamma]^{kN}$ from $\mathsf{H}(\texttt{mask} \,\|\, K \,\|\, i)$
  $w \leftarrow B_1 y$
  $s \leftarrow \mathsf{H}(\texttt{fs} \,\|\, pid \,\|\, |x| \,\|\, x \,\|\, t \,\|\, w)$ ▷ 32 bytes
  $c \leftarrow \mathsf{SampleInBall}(s)$; $z \leftarrow y + cr$
  if $\|z\|_\infty \le B$ **then return** $\pi = (s, \mathrm{Pack}(z + B))$
return $\perp$ ▷ probability $< 2^{-178}$

Verify(pid, x, t, π)
parse $\pi = (s, z)$; reject unless $|\pi|$ is exact, every packed value lies in $[0, 2B]$ and padding bits are zero
$c \leftarrow \mathsf{SampleInBall}(s)$; $w' \leftarrow B_1 z - c\, t_1$
return $s = \mathsf{H}(\texttt{fs} \,\|\, pid \,\|\, |x| \,\|\, x \,\|\, t \,\|\, w')$

TABLE I
PARAMETER SET BDLOP-OPEN-128-V1 AND CORE-SVP ESTIMATES (CRYSTALS SECURITY-ESTIMATES, COMMIT 75c26949).

N, q	256, $2^{30} - 35$ ($q \equiv 5 \bmod 8$)
n, λ, k	6, 6, 13
κ, γ, B	60, 2^{17}, $2^{17} - 61$
Challenge space	2^{257}
Commitment / opening proof	6 720 B / 7 520 B
Acceptance per attempt	0.215
Equality (linkage) proof	15 008 B
Max. in-band payload	766 B
M-SIS ($\ell_\infty \le 8\kappa B \approx 2^{25.9}$)	$\beta = 480$: 140 / 127 bits
M-LWE (rank λ, $n{+}1$ samples)	$\beta = 474$: 138 / 125 bits
β: BKZ block size; core-SVP, classical / quantum.	

two copies of the protocol with a common challenge and adds $u = \langle b_2, y_a - y_b \rangle$ to the hash, which the verifier recomputes as $\langle b_2, z_a - z_b \rangle - c\,(t_{2,a} - t_{2,b})$.

Table I lists the production parameter set. With uniform masking the accepted response is exactly uniform on $[-B, B]^{kN}$, which gives zero knowledge without a Gaussian sampler; the price is a larger extracted norm, so binding is estimated for the ℓ_∞ bound $8\kappa B$ rather than reused from [24]. The per-attempt acceptance probability is $((2B+1)/2\gamma)^{kN} \approx 0.215$ (4.65 expected attempts; 4.68 measured over 50 runs). Core-SVP costs were computed with the CRYSTALS estimation scripts [48], ignoring reduction losses as for ML-DSA; an independent cross-check with the lattice-estimator [49], [50] is **[TODO: cross-check M-SIS/M-LWE estimates with lattice-estimator]**.

E. Envelopes, registry and sessions

The envelope is

$$E_{\text{in}} = \texttt{"ZKLB"} \,\|\, \texttt{0x01} \,\|\, pid \,\|\, N_s \,\|\, t \,\|\, \pi,$$
$$E_{\text{reg}} = \texttt{"ZKLR"} \,\|\, \texttt{0x01} \,\|\, pid \,\|\, N_s \,\|\, \pi,$$

for the in-band and registered modes. In the in-band mode a fresh commitment t (6 720 bytes) to the payload travels in the video; the payload is encoded injectively into N coefficients and is at most 766 bytes. In the registered mode t is the holder's credential commitment, whose bytes enter the verification-key hash that the issuer signs into the registry, so only the credential holder can produce an accepting proof and the video carries only π.

The verifier resolves its pins from the registry: it checks the ML-DSA-65 signature against the pinned issuer key, rejects epochs below its floor, rejects parameter sets with a challenge space below 2^{256}, and requires exactly one record matching its parameter set, carrier profile and (in registered mode) credential. Session challenges are 32-byte random values stored in a SQLite database with a time-to-live and a binding digest over the parameter set, relation, registry root and epoch, codec policy and carrier profile. The verifier reads N_s from the extracted envelope only to look it up; in a single transaction it checks that the challenge was issued, is unexpired, unconsumed and correctly bound, verifies the proof under the statement it built, and marks the challenge consumed only if verification succeeds.

F. Linkage and disclosure

A holder of two openings can prove that two commitments hide the same payload with the equality proof (15 008 bytes), bound to the two commitments and to a context such as the

Algorithm 2 Video-level embedding and verification.

Embed(cover, payload or credential, pins, N_s, key)
$E_0 \leftarrow$ placeholder envelope of the final length; chunk it
$(P, D) \leftarrow$ carrier layout and normalized digest of cover for E_0
$x \leftarrow \mathrm{Statement}(\text{pins}, N_s, |P|, \mathsf{H}(P), V(D))$
$E \leftarrow$ envelope with $\pi \leftarrow$ **Prove**(pid, x, t, r)
abort unless $\mathrm{chunks}(E)$ has the layout of $\mathrm{chunks}(E_0)$
write $\mathrm{chunks}(E)$ into the carriers P by patching; **return** stego

VerifyVideo(stego, pins, session store, key)
$(P', D', E) \leftarrow$ blind extraction (same rule and key); reject on CRC or framing error
$x' \leftarrow \mathrm{Statement}(\text{pins}, N_s(E), |P'|, \mathsf{H}(P'), V(D'))$
return $\mathrm{ConsumeIf}(N_s(E),$ **Verify**$(pid, x', t, \pi))$

two statement identifiers. This links two in-band videos to one payload, or links an in-band video to a registered credential after the fact, without revealing the payload. A holder can also disclose the payload and its randomness to a chosen party, who checks the opening against the commitment extracted from a verified video.

G. Carrier ordering key

With a secret stego key shared by holder and verifier, the carrier layout is unknown to third parties, which is what steganographic undetectability requires: without a secret, anyone who knows the scheme can locate the carriers and test for an envelope [51]. Where undetectability is not required, the ordering key can be derived from public registered configuration, $\mathrm{SHA\text{-}256}(domain \,\|\, pid \,\|\, profile)$, so that anyone with the registry can verify. The key only determines where bits are found: soundness and zero knowledge of the proof do not depend on it, because the verifier recomputes P and V and both enter the statement.

VI. Security analysis

We summarize the argument; full statements and proofs follow the published analyses of BDLOP [24] and ML-DSA [12], [31] with uniform masking.

Theorem 1 (informal)**.** *In the classical random-oracle model, the opening proof is complete with per-attempt acceptance $((2B+1)/2\gamma)^{kN}$; it is honest-verifier zero knowledge for accepted transcripts; and it is special sound, with extractor output a relaxed opening $(\bar{r}, f)$ satisfying $B_1\bar{r} = f\,t_1$, $\|\bar{r}\|_\infty \le 2B$ and f a difference of two challenges, hence invertible. Two relaxed openings of one commitment to different messages yield a Module-SIS solution for B_1 of ℓ_∞ norm at most $8\kappa B$; hiding reduces to Module-LWE of rank λ with $n+1$ samples and ternary secret and error.*

Knowledge soundness of the non-interactive proof follows from special soundness by the forking lemma [52], [53], with a non-tight reduction; the rejection-sampling bias in the random-oracle reprogramming is handled as in [44], [45]. The parameter identifier, the 274-byte statement and the full commitment are hashed before the challenge is derived, so a proof verifies only for the statement it was made for; since the verifier recomputes every statement field, the proof is bound to the carrier-normalized video, the carrier layout, the registry entry and the session (G1, and G3 for the envelope bytes), and the atomic consumption of N_s gives G4. The equality proof yields relaxed openings with a common f, from which the two extracted messages are equal (G5). We call the scheme *plausibly post-quantum*: the underlying problems have no known efficient quantum algorithm, but we do not claim a proof in the quantum random-oracle model.

VII. Implementation

The backend is a self-contained, pure-Python implementation (Python 3.12, NumPy) of about 1 700 lines for the ring, sampling, commitment, opening and equality proofs, payload codec, statement, envelope, linkage and a command-line interface that separates the issuer, holder and verifier roles. Polynomial multiplication uses exact big-integer Kronecker substitution; there is no NTT because $q \equiv 5 \pmod 8$ does not split $X^N + 1$ fully. The H.264 side reuses a Python CAVLC parser and bitstream patcher with a blind carrier synchronizer, a carrier canonicalizer, the chunk framing, an ML-DSA-65-signed relation registry (via the `pqcrypto` bindings) and the SQLite session store. The test suite covers the ring and samplers, commitment and proof correctness, the accepted-response uniformity and the acceptance edge, extraction of relaxed openings from two transcripts, statement substitution, envelope tampering, registry pinning and rollback, session replay, linkage, the CLI, and byte-exact known-answer vectors for the parameter set.

a) Sizes: Commitment 6 720 B, opening proof 7 520 B (a 32-byte challenge seed and kN responses of 18 bits), equality proof 15 008 B, envelope header 69 B; hence 7 589 B (registered) and 14 309 B (in-band) in the video, plus 23 B of framing per chunk.

b) Timing: On the evaluation host (single thread, idle), the median times over 50 runs are 21 ms to commit, 120 ms to prove (95th percentile 302 ms) and 22 ms to verify an opening, and 0.97 s and 46 ms for the equality proof. Cryptography is therefore negligible end to end: embedding and verification of a clip take minutes and are dominated by the Python CAVLC analysis and patching (Table III). A faster verification path is in progress; **[TODO: update verification times after the verification speed-up (roadmap A2)]**.

c) Why not a native encoder path: Embedding inside an encoder would change the reconstructed reference pixels, and with them all subsequent intra-predicted values, as a function of the proof bits, so a statement could not bind the video without a codec-level relation. Post-encoding patching with a carrier-normalized digest keeps the statement independent of the proof.

VIII. Evaluation

A. Setup

All measurements were taken on one Windows 11 host with an Intel Core i7-12700H (14 cores) and 24 GB of RAM. Sources are standard Xiph test sequences: CIF (352×288; 300 frames, except Deadline with 1 374 and Football with 260) and 4CIF (704×576; 600 frames). They were encoded

TABLE II
STABLE, PATCHABILITY-CHECKED MAGNITUDE CARRIERS (ALL-INTRA, X264 BASELINE/CAVLC). SEG. SCANNED: 100-FRAME SEGMENTS MEASURED OUT OF THE CLIP'S TOTAL; UNSCANNED SEGMENTS ARE EXTRAPOLATED FROM THE SCANNED ONES. FRAMES NEEDED: SEGMENTS REQUIRED AT THE SMALLEST FEASIBLE CHUNK SIZE; † MORE THAN THE CLIP HAS.

Clip	Res.	QP	Frames	Seg.	kbit/100 fr.		Frames needed	
				scanned	min	mean	reg.	in-band
Akiyo	CIF	22	300	1/3	10.5	10.5	600†	1200†
Foreman	CIF	22	300	1/3	14.1	14.1	500†	900†
Football	CIF	22	260	3/3	14.3	21.6	500†	900†
City	CIF	22	300	3/3	21.7	23.8	300	600†
Hall	CIF	22	300	3/3	27.1	27.6	300	500†
Coastguard	CIF	22	300	3/3	31.4	36.3	200	400†
Deadline	CIF	22	1374	1/14	46.6	46.6	200	300
Container	CIF	22	300	3/3	47.9	49.8	200	300
Container	CIF	28	300	3/3	9.1	10.0	700†	1300†
Container	CIF	18	300	3/3	86.3	88.3	100	200
Crew	4CIF	22	600	3/6	11.1	13.2	600	1100†
Soccer	4CIF	22	600	3/6	3.5	30.0	1900†	3600†
Harbour	4CIF	22	600	4/6	48.8	50.0	200	300
City	4CIF	22	600	3/6	68.1	82.3	100	200

with FFmpeg/libx264 in the Baseline profile (CAVLC), all-intra ($GOP = 1$, no B-frames) at a fixed QP of 18, 22 or 28. Two longer inputs were used for the earliest runs: the first 1 000 frames of Deadline, and Coastguard looped ten times to 3 000 frames (marked $*$). Quality is the FFmpeg luma PSNR of the decoded stego against the decoded cover; the PSNR of modified frames is computed from their mean MSE, and SSIM is over the whole video. Unless stated otherwise the registered mode and the magnitude carrier are used, with 100-frame segments and a chunk size chosen from the measured capacity of each clip. Experiments ran sequentially, except where noted in the tables.

B. Capacity

Table II lists stable, patchability-checked carriers. At QP 22 CIF clips offer 10.5–49.8 kbit per 100 frames. Textured content (Container, Deadline, Coastguard) carries the registered envelope (about 61 kbit) in 200 frames, and Container and Deadline carry the in-band envelope (about 115 kbit) in 300; flat or smooth content (Akiyo, Foreman) cannot carry the registered envelope in its 300 frames. Capacity falls steeply with QP (Container: 88.3, 49.8 and 10.0 kbit at QP 18, 22 and 28), because coarser quantization leaves fewer coefficients with $|level| \geq 4$, and grows with resolution where the content is textured (City: 23.8 kbit at CIF, 82.3 kbit at 4CIF). Capacity also varies strongly within a clip: one Soccer segment offers only 3.5 kbit, and Crew offers 11–17 kbit per segment, so the sweep skipped both 4CIF clips as not fitting. The capacity scan exposed a defect in the carrier analysis at non-CIF resolutions, which mapped macroblocks to frames with the CIF macroblock count and so discarded most of each 4CIF frame; it failed closed (no wrong data, only lost capacity) and was fixed before the 4CIF runs.

C. End-to-end embedding and video-only verification

Table III reports the end-to-end runs. Every completed run verified from the stego video alone, and in the runs that exercised them a replay of the consumed session and a copy with one flipped bit were both rejected; the payload or credential never appears in the output file. The in-band runs additionally verified a linkage proof between the commitment extracted from the video and a separately committed credential, rejected the linkage to a different credential, and verified a later disclosure of the payload. On modified frames the luma PSNR is 40.0–45.4 dB, the worst frame is 32.1–39.3 dB, and whole-video SSIM is at least 0.9991. The worst frames are those in which many carriers fall in smooth regions, where intra prediction spreads a ± 1 change to neighbouring blocks. Embedding takes 6–63 minutes (90 minutes for the concurrently run 3 000-frame in-band case) and verification 2–33 minutes per clip in pure Python; the proof itself takes about 0.1 s. Pending runs are marked in the table: **[TODO: City 4CIF at QP 18 and 28]**.

D. Steganalysis

a) Classical: Targeted tests against the exact cover, on the first 100 frames of each stego video, do not separate cover and stego (Table IV): the Westfeld–Pfitzmann pairs-of-values attack on carrier magnitudes [21] returns an embedding probability of zero for both cover and stego, the two-sample magnitude histogram and sign tests give $p \geq 0.998$, and SPA on decoded luma [22] changes in the fourth decimal. RS analysis returned no difference for cover or stego on this content and is not used as evidence.

b) Learned: A learned detector is more effective. We extract, per frame, 340 coefficient-domain features (per-position magnitude histograms, odd-magnitude rates for $|level| \geq 4$, adjacent-position magnitude co-occurrences and per-position sign shares) and 500 pixel-domain features from a reduced spatial rich model [18] (quantized first- and second-order residual co-occurrences), and train logistic regression and an ensemble of shrinkage Fisher linear discriminants on random 30 % feature subspaces (51 members), in the spirit of [19]. The attacker is strong: it trains on exact cover/stego pairs from the same encoder settings and knows which frames carry bits; only modified frames are used. We report the minimum average error $P_E = \min(P_{\mathrm{FA}} + P_{\mathrm{MD}})/2$. With leave-one-clip-out evaluation over five CIF clips (1 300 frame pairs), the best detector reaches $P_E = 0.40$ with coefficient features and 0.45 with pixel features; with clips seen in training (a random split of pairs), the combined features reach 0.36 (Table V). The embedding is thus detectable per frame at a moderate level in its own domain and barely in the pixel domain. These are per-frame figures; a detector that pools the evidence of all modified frames of a video [23] would do better (Section IX). A CNN detector [20] trained on the same protocol is **[TODO: CNN steganalysis P_E (roadmap A6)]**, and the learned evaluation over all clips including 4CIF is **[TODO: final ML evaluation including 4CIF clips]**.

TABLE III

END-TO-END RUNS WITH THE MAGNITUDE CARRIER. EVERY COMPLETED RUN WAS VERIFIED FROM THE STEGO VIDEO ALONE. PSNR IS LUMA PSNR (DB) AGAINST THE DECODED COVER, COMPUTED FROM THE MEAN MSE OVER MODIFIED FRAMES (MOD), THE WORST FRAME (MIN) AND ALL FRAMES (ALL); SSIM IS OVER THE WHOLE VIDEO. CHECKS: R = REPLAY OF THE CONSUMED SESSION REJECTED, T = A ONE-BIT CHANGE OF THE STEGO FILE REJECTED, L = LINKAGE PROOF TO A CREDENTIAL VERIFIED AND A DIFFERENT CREDENTIAL REJECTED, D = LATER DISCLOSURE OF THE PAYLOAD VERIFIED; – = NOT EXERCISED IN THAT RUN. * 300-FRAME SOURCE LOOPED TO 3 000 FRAMES; THESE TWO RUNS WERE EXECUTED CONCURRENTLY, SO THEIR TIMES ARE INFLATED.

Clip	Res.	QP	Mode	Frames	Seg.	Carriers	Mod. fr.	$\mathrm{PSNR}_{\mathrm{mod}}$	$\mathrm{PSNR}_{\mathrm{min}}$	$\mathrm{PSNR}_{\mathrm{all}}$	SSIM	Embed (s)	Verify (s)	Verified	Checks
City	CIF	22	reg.	300	3	61 600	300	45.35	38.81	45.35	0.99969	903	286	yes	–
Coastguard	CIF	22	reg.	300	2	61 304	200	39.97	33.48	41.73	0.99958	702	271	yes	–
Container	CIF	18	reg.	300	1	61 008	100	43.94	36.41	48.71	0.99989	334	127	yes	–
Container	CIF	22	reg.	300	2	61 304	200	42.07	34.02	43.83	0.99975	580	182	yes	R,T
Deadline	CIF	22	reg.	1000	7	62 784	684	44.99	34.19	46.64	0.99984	1793	1261	yes	R,T
Hall	CIF	22	reg.	300	3	61 600	300	41.48	32.72	41.48	0.99911	613	238	yes	–
Coastguard*	CIF	22	reg.	3000	7	62 784	694	45.26	35.66	51.61	0.99996	2164	740	yes	R,T
City	4CIF	22	reg.	600	1	61 008	100	43.66	39.31	51.44	0.99995	881	584	yes	R,T
Harbour	4CIF	22	reg.	600	2	61 304	200	42.83	32.91	47.60	0.99994	2357	1954	yes	–
Container	CIF	18	in-band	300	2	115 064	200	44.12	38.42	45.88	0.99980	678	465	yes	L,D
Container	CIF	22	in-band	300	3	115 360	300	40.26	32.13	40.26	0.99949	719	466	yes	R,T,L,D
Coastguard*	CIF	22	in-band	3000	12	118 024	1200	45.10	36.02	49.08	0.99992	5409	1707	yes	R,T,L,D
Harbour	4CIF	22	in-band	600	3	115 360	300	41.16	33.15	44.17	0.99987	3760	1108	yes	L,D
Crew	4CIF	22	reg.	skipped by the sweep: envelope does not fit the scanned capacity											
Soccer	4CIF	22	reg.	skipped by the sweep: envelope does not fit the scanned capacity											
City	4CIF	18	reg.	**[TODO]** pending run											
City	4CIF	28	reg.	**[TODO]** pending run											

TABLE IV

TARGETED CLASSICAL STEGANALYSIS AGAINST THE EXACT COVER (MAGNITUDE CARRIER). POV: WESTFELD–PFITZMANN EMBEDDING PROBABILITY ON CARRIER MAGNITUDES FOR COVER/STEGO; MAG., T1, POS.: p-VALUES OF TWO-SAMPLE TESTS ON THE MAGNITUDE HISTOGRAM, TRAILING-ONE SIGNS AND PER-SCAN-POSITION SIGNS ($p < 0.05$ WOULD INDICATE DETECTION); SPA: SAMPLE-PAIR ESTIMATE ON DECODED LUMA FOR COVER/STEGO.

Clip	Res.	QP	Mode	Frames	PoV P (c/s)	Mag. p	T1 p	Pos. p	SPA (c/s)
City	4CIF	22	reg.	100	0.000 / 0.000	0.998	1.000	1.000	0.3945 / 0.3946
City	CIF	22	reg.	100	0.000 / 0.000	1.000	1.000	1.000	0.4076 / 0.4077
Coastguard*	CIF	22	reg.	100	0.000 / 0.000	1.000	1.000	1.000	0.4066 / 0.4066
Coastguard	CIF	22	reg.	100	0.000 / 0.000	1.000	1.000	1.000	0.4066 / 0.4067
Container	CIF	18	in-band	100	0.000 / 0.000	1.000	1.000	1.000	0.3314 / 0.3316
Container	CIF	18	reg.	100	0.000 / 0.000	1.000	1.000	1.000	0.3314 / 0.3316
Container	CIF	22	in-band	100	0.000 / 0.000	1.000	1.000	1.000	0.2771 / 0.2775
Container	CIF	22	reg.	100	0.000 / 0.000	1.000	1.000	1.000	0.2771 / 0.2773
Deadline	CIF	22	reg.	100	0.000 / 0.000	1.000	1.000	1.000	0.2879 / 0.2881
Hall	CIF	22	reg.	100	0.000 / 0.000	1.000	1.000	1.000	0.2645 / 0.2654
Harbour	4CIF	22	in-band	100	0.000 / 0.000	1.000	1.000	1.000	0.4242 / 0.4243
Harbour	4CIF	22	reg.	100	0.000 / 0.000	1.000	1.000	1.000	0.4242 / 0.4244

TABLE V

LEARNED PER-FRAME STEGANALYSIS OF THE MAGNITUDE CARRIER ON 1 300 MODIFIED COVER/STEGO FRAME PAIRS FROM 5 CIF CLIPS ($P_E = 0.5$ IS CHANCE). RANGE: MINIMUM AND MAXIMUM P_E OVER THE HELD-OUT CLIPS.

Features	Detector	Leave-one-clip-out			Paired split	
		P_E	range	AUC	P_E	AUC
Coefficient (340)	logistic	0.407	0.35–0.47	0.604	0.414	0.617
	FLD ens.	0.399	0.34–0.45	0.628	0.394	0.638
Pixel, reduced SRM (500)	logistic	0.454	0.41–0.48	0.538	0.424	0.608
	FLD ens.	0.458	0.42–0.49	0.536	0.410	0.627
Combined (840)	logistic	0.437	0.41–0.47	0.575	0.401	0.633
	FLD ens.	0.422	0.40–0.45	0.583	0.357	0.685

E. Carrier choice

We compared the magnitude carrier with the trailing-one sign carrier used by earlier CAVLC schemes [15], through the same pipeline, on four CIF clips, both at the same number of modified frames as the magnitude carrier and packed into 100 frames (Table VI). All twelve runs verify. Sign carriers are an order of magnitude more plentiful (468–550 kbit per 100 frames). At equal frame count they give higher PSNR on modified frames (43.3–48.8 vs. 40.0–45.4 dB; worst frame 38.5–44.6 vs. 32.7–38.8 dB) but slightly lower whole-video SSIM, and they are much harder to see in the coefficient domain (leave-one-clip-out $P_E = 0.47$ vs. 0.39) yet easier in the pixel domain (0.40 vs. 0.46); with combined features the two are close (0.40 vs. 0.42). Packed into 100 frames, sign flips are clearly detectable ($P_E = 0.31$ with pixel features, and a per-position sign test flags one clip at $p = 0.001$). Sign carriers were 4–8 times slower to verify at equal frame count (2–3 times when packed into 100 frames), because the reduced stable-carrier analysis does not apply to them; **[TODO: timing of the optimized sign-carrier analysis (t1-sign-video-only-v2) once measured on an idle host]**. We keep the magnitude carrier as the default for verification speed and pixel-domain behaviour, and expose the sign carrier as an option when visual quality matters more.

TABLE VI
CARRIER COMPARISON THROUGH THE SAME PIPELINE ON FOUR CIF CLIPS (CITY, COASTGUARD, CONTAINER, HALL; QP 22, REGISTERED MODE). RANGES ARE OVER THE FOUR CLIPS. "SAME FRAMES" USES THE CHUNK SIZE OF THE MAGNITUDE RUN, SO BOTH CARRIERS TOUCH THE SAME NUMBER OF FRAMES; "100 FRAMES" PACKS THE ENVELOPE INTO ONE SEGMENT. MIN POS. p: SMALLEST PER-POSITION SIGN-TEST p-VALUE OVER THE CLIPS.

Carrier	Verified	kbit/100 fr.	PSNR_{mod}	PSNR_{min}	SSIM	Embed (s)	Verify (s)	min Pos. p	Learned P_E			
									Coef. LOCO	Pixel LOCO	Comb. LOCO	Comb. paired
Magnitude LSB	4/4	21.7–47.9	40.0–45.4	32.7–38.8	0.9991–0.9998	580–903	182–286	1.000	0.386	0.456	0.416	0.339
T1 sign, same frames	4/4	467.6–549.7	43.3–48.8	38.5–44.6	0.9975–0.9994	1275–2329	788–2395	0.981	0.471	0.404	0.401	0.317
T1 sign, 100 frames	4/4	467.6–549.7	41.0–44.1	36.8–40.6	0.9978–0.9994	857–1338	519–816	0.001	0.446	0.310	0.314	0.193

TABLE VII
DETECTABILITY VERSUS EMBEDDING RULE AND DENSITY: LEAVE-ONE-CLIP-OUT, COEFFICIENT FEATURES, FLD ENSEMBLE, FIVE CIF CLIPS. THE UPPER ROWS ARE SIMULATED ON THE REAL CARRIER POSITIONS WITHOUT THE PATCHABILITY FILTER (HENCE MORE DETECTABLE THAN THE REAL RUNS, $P_E = 0.40$); THEY ARE FOR RELATIVE COMPARISON ONLY.

Embedding (fraction of carriers used)	P_E	AUC
Magnitude LSB replacement, 100 %	0.315	0.737
Magnitude LSB replacement, 50 %	0.395	0.627
Magnitude LSB replacement, 25 %	0.443	0.566
Magnitude LSB replacement, 10 %	0.466	0.527
Magnitude LSB matching (± 1), 100 %	0.267	0.801
T1 sign flip, 100 %	0.474	0.512
Measured 4CIF E2E, 100 %	**[TODO]**	**[TODO]**
Measured 4CIF E2E, 50 %	**[TODO]**	**[TODO]**
Measured 4CIF E2E, 25 %	**[TODO]**	**[TODO]**

F. Embedding rule and density

Table VII uses simulated embedding on the real carrier positions of the five CIF clips to compare rules and densities with the same detector. LSB matching (± 1 chosen at random) is *more* detectable than replacement ($P_E = 0.27$ vs. 0.32), so replacement is kept. Using 50, 25 and 10 % of the carriers raises P_E to 0.40, 0.44 and 0.47; the envelope can be spread by lowering the chunk size, at the cost of touching more frames. The simulation ignores the patchability filter and is therefore more detectable than the real runs; it supports relative comparisons only. Measured density results on the 4CIF clips are **[TODO: measured 4CIF density curve at 100/50/25 %]**.

G. Comparison with alternative proof systems

Table VIII compares schemes that could carry the claim. Groth16 is about fifty times smaller but not post-quantum and needs a circuit-specific trusted setup. ML-DSA signatures are post-quantum and smaller, but a signature reveals the signed data or binds a public value; it proves nothing about a hidden payload and supports neither later linkage nor selective disclosure. ISW21 is a lattice zkSNARK with a designated verifier, statistical zero knowledge of only 2^{-40} in the configuration we ran, a 2.4 MB verifier key, and an analysis based on an assumption invalidated against quantum adversaries [38]; its envelope is also larger. The BDLOP opening proof is the smallest transparent, publicly verifiable, zero-knowledge option we found that fits the measured capacity.

IX. DISCUSSION AND LIMITATIONS

a) Relaxed soundness: The extractor yields a relaxed opening $(\bar{r}, f)$ rather than a ternary opening, so the payload is bound only through binding with respect to relaxed openings. This is standard for Fiat–Shamir-with-aborts proofs, but the statement "the holder knows the payload" must be read in this relaxed sense.

b) Random-oracle model: The argument is in the classical random-oracle model with a non-tight forking reduction. We do not claim security in the quantum random-oracle model; generic results for Fiat–Shamir [46], [47] make a post-quantum interpretation plausible but do not give concrete bounds for our parameters.

c) Implementation: The code is a research prototype implemented by us from the published protocol. It has not been audited and is not constant time (Python big-integer arithmetic and data-dependent rejection loops); timing side channels are out of scope.

d) Framing malleability: The envelope bytes are integrity-protected: changing any of them makes the proof or the statement fail. The chunk framing around them is not fully bound. When the envelope fits in a single chunk, up to about 48 carrier bits of the chunk header (including the redundant chunk-size field) can currently be changed without the video being rejected; the extracted envelope, and hence what is verified, is unchanged, but the stego video is then not unique. Binding the header into the CRC or the statement is a pending fix; **[TODO: update after the chunk-framing fix is merged]**.

e) Stego key versus public order: Undetectability requires a symmetric key shared with the verifier, which makes verification designated in practice. The public-order mode makes verification public and gives up undetectability. The keyed permutation that orders carriers is derived from an HMAC-SHA-256 seed but uses a chaotic-map shuffle, which is not a cryptographic pseudorandom permutation; replacing it with a standard keyed permutation is straightforward and does not affect the proof.

f) Per-frame versus per-video detection: The learned detector's per-frame error of 0.40 means that each modified frame leaks a little; a video carries 100–1 200 modified frames, and pooling their scores would likely detect the embedding with high confidence. We have not quantified a pooled detector. Spreading the envelope over more frames lowers per-frame density but increases the number of frames that leak, so cost-aware carrier selection is the more promising direction.

TABLE VIII
PROOF AND SIGNATURE SCHEMES THAT COULD CARRY THE CLAIM, ON THE SAME HOST (SINGLE THREAD). SIZE: BYTES CARRIED IN THE VIDEO (ENVELOPE OR PROOF-BEARING PAYLOAD); THE EQUALITY PROOF IS EXCHANGED OUT OF BAND AT LINKAGE TIME. BDLOP TIMINGS ARE MEDIANS OF 50 RUNS ON AN IDLE MACHINE (PURE PYTHON); GROTH16 IS END-TO-END WALL TIME THROUGH A PYTHON BRIDGE (3 TRIALS); ML-DSA VIA PQCRYPTO. ISW21 NUMBERS WERE RECORDED EARLIER IN A LINUX CONTAINER ON THE SAME CPU (TOY 32-BYTE RELATION) AND NOT RE-RUN. ‡ THE OBLIVIOUS-LWE-SAMPLING ASSUMPTION USED IN ITS ANALYSIS IS INVALIDATED AGAINST QUANTUM ADVERSARIES [38].

Scheme	Assumption	ZK	Hides payload	Size (B)	Prove (ms)	Verify (ms)	Setup
This work, registered	M-SIS/M-LWE	yes	yes	7589	120	22	transparent
This work, in-band	M-SIS/M-LWE	yes	yes	14 309	120	22	transparent
Equality (linkage) proof	M-SIS/M-LWE	yes	–	15 008	968	46	transparent
Groth16 (BN128)	none (pairings)	yes	yes	147	7222	2550	per-circuit trusted
ML-DSA-44 signature	M-LWE/M-SIS	no	no	2420	18.6	0.33	none
ML-DSA-65 signature	M-LWE/M-SIS	no	no	3309	25.3	0.52	none
ISW21 (recorded)	lattice + knowledge‡	stat. 2^{-40}	yes	23 907	328	0.6	trusted, designated

g) Capacity and codec scope: Only all-intra Baseline streams are supported, which is a real deployment restriction. A registered envelope needs 61 kbit, which smooth content and high QP do not provide within a few hundred frames. Proving that each payload bit is binary, or that the payload lies in a range, would need a fully splitting ring and the product proofs of [27], [28], which this parameter set does not provide.

h) What is claimed: An accepted video shows that its producer knew a (relaxed) opening of the bound commitment for this exact carrier-normalized video, carrier layout, registry entry and session; in registered mode, that the producer holds the registered credential. It does not show that the payload is true, that a camera recorded the video, or, in in-band mode, who produced it. The session challenge is not secret, so it should be delivered over an authenticated channel if the holder's identity matters.

X. THREATS TO VALIDITY

a) Internal validity: The protocol, parser and patcher were implemented by the authors; correctness is supported by unit and negative tests, known-answer vectors and the end-to-end verification itself, but not by an independent implementation. Quality is measured against the decoded cover, not the raw source, which isolates the embedding from the encoder. Timings come from a single host; two long runs were executed concurrently and are marked. The density and rule comparisons are simulations that omit the patchability filter.

b) Construct validity: P_E is a per-frame measure on modified frames only, which favours the attacker within a frame but ignores pooling across frames. PSNR on modified frames depends on how many frames an envelope touches, which in turn depends on the chosen chunk size. Capacity for partly scanned clips is extrapolated from the scanned segments.

c) External validity: The evaluation uses a small set of standard CIF and 4CIF sequences, one encoder (x264), the Baseline profile, all-intra coding and fixed QP. Results may differ for camera footage, other encoders, rate control, P-slices or higher resolutions. The detectors are feature-based; a stronger learned detector may lower P_E.

d) Conclusion validity: Learned results are averaged over five held-out clips with no confidence intervals; we report the range over folds instead. With few clips, differences of a few hundredths in P_E between configurations should not be over-interpreted.

XI. CONCLUSION

A post-quantum zero-knowledge proof can travel inside an H.264 CAVLC bitstream, bound to that bitstream through a carrier-normalized commitment and to a verifier session, and be verified from the video alone. With a 7.6 kB registered envelope and a 14.3 kB in-band envelope, textured CIF and 4CIF content carries the proof in one to three hundred frames at 40–45 dB on modified frames, and classical targeted steganalysis does not detect it. A learned per-frame detector does better than chance, and the security argument rests on relaxed soundness in the classical random-oracle model. The next steps are cost-aware carrier selection against learned and pooled detectors, binary and range proofs on a fully splitting ring, support for P-slices and higher resolutions, and an independent review of the cryptographic argument.

ARTIFACT AVAILABILITY

The implementation, the specification with known-answer vectors, the parameter-estimation scripts, all result files used in this paper and the script that regenerates its tables are available in an anonymized repository: **[TODO: anonymized repository URL]**. Test videos are derived from the public Xiph sequences with the provided encoding commands; the large intermediate files are not redistributed.

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